POKHARA UNIVERSITY

Level: Bachelor Semester:Fall Year : 2020
Programme:BE Full Marks: 100
Course:Chemistry Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define buffer solution. Explain the mechanism of basic buffer. Calculate the pH of a buffer solution prepared by mixing 500ml of 0.6M sodium acetate and 700ml of 0.3M acetic acid. 8

   b) Write the mechanism of chemical and electrochemical corrosion. From the given electrode potential values answer the following : 7

   i. Write the electrode reaction.

   ii. Write the cell reaction.

   iii. Calculate the e.m,f. of the cell at 298K when both the electrodes are coupled together.

   $E^0\ Zn^{2+}/Zn = -0.76V$

   $E^0\ Cu^{2+}/Cu = +0.34\ V$

   $R = 8.314\ J\ mol^{-1}\ K^{-1}$

   $F = 96500C$

   $[Zn^{2+}] = 0.1M$ $[Cu^{2+}] = 1.75M$

2. a) Give reason: 8

   i. Transition elements are good in forming complexes.

   ii. Ionization energy of neon is higher than fluorine.

   iii. TiO_2 is white but $TiCl_3$ is violet.

   iv. Cu(I) is diamagnetic whereas Cu (II) is paramagnetic.

   b) What is electron affinity? Write down the factors that affects the magnitude of electron affinity. Why is electron affinity of chlorine higher than fluorine? 7

3. a) What do you mean by SN_1 and SN_2 reaction? Explain the mechanism with specific example. 8

   b) How do enantiomers differ from diasteriomers? Explain the stability of free radicals. 7

4. a) What are polymers? Discuss about the mechanism of condensation polymerization taking a suitable example. 8

   b) What is IUPAC name for monomer of teflon? Show the free radical addition mechanism for the preparation of teflon? 7

5. a) Write the principle of Chromatography. Write the important applications of thin layer chromatography. 8

   b) What is cement? Why gypsum is added in the cement? Write the method of preparations of cement in brief. 7

6. a) Define acid rain. Discuss about the sources, effects and important control methods of acid rain. 8

   b) Define hardness of water. How it can be measured in the laboratory? Describe it. 7

7. Write short notes on: **(Any two)** 2×5

   a) Mass Spectroscopy.

   b) Lubricants and its types.

   c) Green house effect